

Section 4.3

The Internet Protocol (IP):
IPv4, Addressing, IPv6, and More

Kurose & Ross, 8th Ed.



4.3 — The Internet Protocol — Roadmap

Section 4.3: Introduction

IP is the universal language of the Internet — every packet ever sent carries an IP header

4.3.1 IPv4 Datagram Format

The 20-byte header: Version, TTL, Protocol, Checksum, Src/Dst addresses, fragmentation fields. Overhead analysis.

4.3.2 IPv4 Addressing

Interfaces and IP addresses. Subnets and /24 notation. CIDR — Classless Interdomain Routing. Route aggregation. DHCP: plug-and-play.

4.3.3 NAT

How billions of devices share a handful of public IP addresses. NAT translation table. Port reuse. Controversy.

4.3.4 IPv6

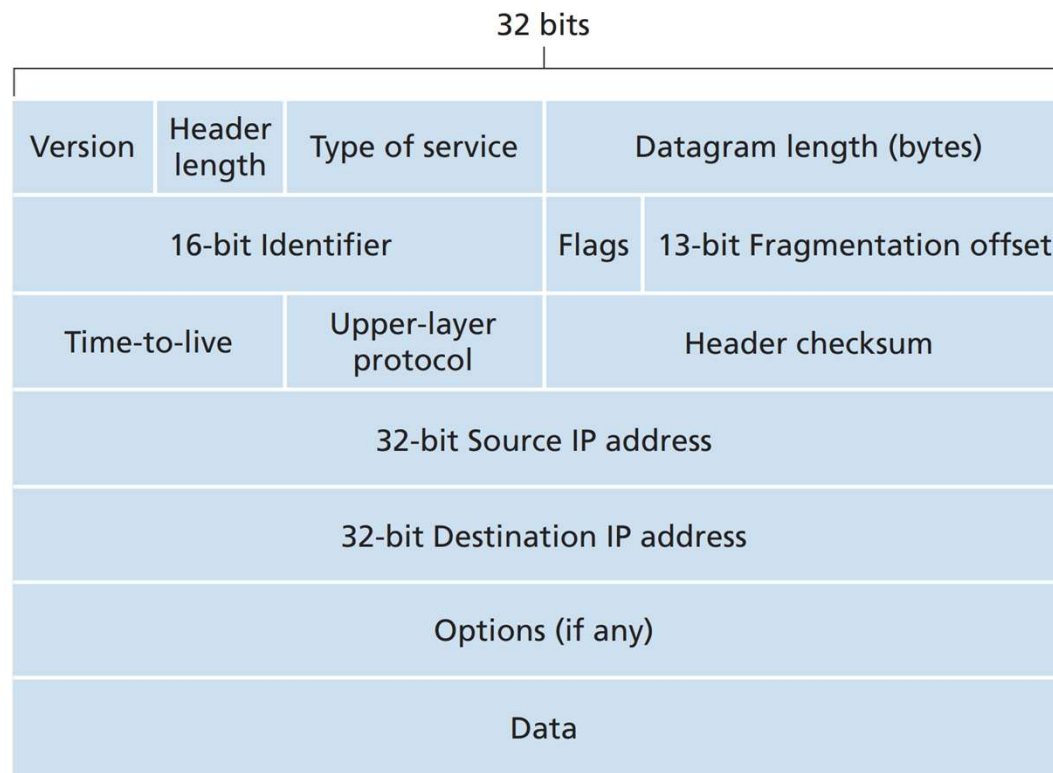
128-bit addresses. Fixed 40-byte header. Fields removed vs IPv4 (fragmentation, checksum, options). Flow Label. IPv4→IPv6 tunneling.

To master IP addressing is to master the Internet's network layer itself.

4.3.1 IPv4 Datagram Format

Section 4.3.1: IPv4 Datagram Format

The Internet's network-layer packet is called a **DATAGRAM** — minimum 20-byte header, variable data payload



Typical IPv4 datagram: 20 bytes IP header + 20 bytes TCP header = 40 bytes overhead before application data begins

4.3.1 IPv4 Datagram Format

Section 4.3.1: IPv4 Datagram Format

Field	Size	Description
Version	4 bits	IP version number. IPv4 = 4. Routers use this to interpret the rest of the header.
Header Length	4 bits	Where payload begins (header has variable-length options). Min = $5 \times 32 = 20$ bytes.
Type of Service	8 bits	Mark datagrams for priority (e.g., VoIP vs bulk transfer). 2 bits used for ECN.
Datagram Length	16 bits	Total header + data in bytes. Max = 65,535. Rarely > 1,500 bytes (fits one Ethernet frame).
ID / Flags / Offset	32 bits	Fragmentation fields: identify, flag (more fragments?), and locate fragments at destination.
TTL	8 bits	Decrement at each router. Dropped when 0. Prevents routing loops. Powers Traceroute.
Protocol	8 bits	Identifies transport-layer protocol at destination: 6=TCP, 17=UDP, 1=ICMP.
Header Checksum	16 bits	1's complement sum over header fields. Recomputed at EVERY router (TTL changes each hop).

IPv4 Header — Key Fields in Depth

Section 4.3.1: IPv4 Datagram Format

TTL — Time to Live

Decrement by 1 at EACH ROUTER.
Dropped at 0 → router sends ICMP "Time Exceeded" error to source.

Why: prevents datagrams from circulating forever due to routing loops.

Bonus: enables TRACEROUTE!
Probes with TTL = 1, 2, 3... force successive routers to discard and reveal their identity.

Typical initial TTL: 64 (Linux) or 128 (Windows).

Protocol — Demux Key

Used ONLY at the final destination.
Tells network layer which transport protocol receives the payload:

6 → TCP
17 → UDP
1 → ICMP

The GLUE binding network layer to transport layer — same role as port numbers binding transport to application.

Example: Protocol = 6 → pass payload to TCP.

Fragmentation

Large datagrams crossing a small-MTU link are SPLIT.

Example: 4,000 B → three fragments on 1,500 B Ethernet link.

Each fragment: same Identifier, sequential Offset, More-fragments flag.

Reassembly at DESTINATION ONLY — not at intermediate routers.

IPv6 removes fragmentation entirely from routers — source is responsible.

Overhead example: app sends 40-byte chunks every 20 ms (VoIP codec) → 40-byte IP+TCP header per datagram → 50% of each datagram is pure overhead.

Source and Destination IP addresses (32 bits each) identify the sender and intended recipient of every datagram

4.3.2 IPv4 Addressing — Interfaces

Section 4.3.2: IPv4 Addressing

An IP address is associated with an INTERFACE — not with a host or router

What Is an Interface?

The boundary between a host (or router) and the physical link connecting it to the network.

Every host and router INTERFACE must have its own globally unique IP address.

A LAPTOP has one interface → one IP address.

A ROUTER has multiple interfaces (one per connected link) → MULTIPLE IP addresses. A router with 3 links has 3 IP addresses.

Interfețele point-to-point between routers are also separate IP addresses.

IPv4 Address Format

32 bits long = $2^{32} \approx 4.3$ billion possible addresses.

Written in DOTTED-DECIMAL NOTATION:

Four bytes, each 0–255, separated by dots.

Example:

193.32.216.9

= 11000001.00100000.11011000.00001001

Two interfaces on the SAME SUBNET share a common prefix in their addresses.

Example: hosts 223.1.1.1, 223.1.1.2, 223.1.1.3 are on the 223.1.1.0/24 subnet.

An IP address technically belongs to an INTERFACE. This is why a router has a different IP on each of its links.

Subnets — Isolating Network Segments

Section 4.3.2: IPv4 Addressing

A SUBNET: set of interfaces that can reach each other **WITHOUT** passing through an intervening router

Subnet Notation and Subnet Mask

Subnet address written as: a.b.c.d/x
The /x (slash-x, or subnet mask) means the leftmost x bits define the SUBNET. The remaining 32-x bits identify hosts.

Example: 223.1.1.0/24

- Leftmost 24 bits: 223.1.1 → subnet address
- Remaining 8 bits: identify up to $2^8 - 2 = 254$ hosts
- Hosts: 223.1.1.1 ... 223.1.1.254
- Reserved: 223.1.1.0 (network) and 223.1.1.255 (broadcast)

How to IDENTIFY SUBNETS:

Detach each interface from its host/router. Each isolated island of network = one subnet.

Example: 3 Routers — 6 Subnets

Three routers interconnected, each with 3 interfaces.

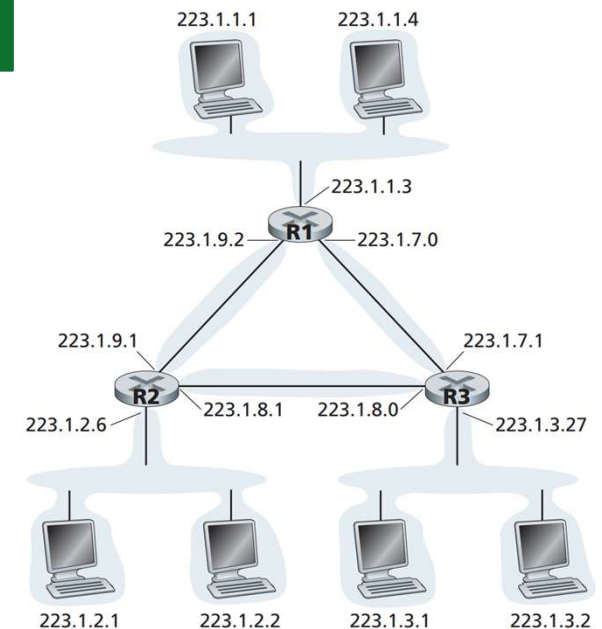
Subnets with HOSTS (3):

- 223.1.1.0/24 — top
- 223.1.2.0/24 — bottom-left
- 223.1.3.0/24 — bottom-right

Subnets on POINT-TO-POINT LINKS (3):

- 223.1.7.0/24 — R1 ↔ R3
- 223.1.8.0/24 — R2 ↔ R3
- 223.1.9.0/24 — R1 ↔ R2

Total: 6 subnets.
Each router interface has its own IP address.



Point-to-point links between routers are also subnets — each router interface has its own IP address

CIDR — Classless Interdomain Routing

Section 4.3.2: IPv4 Addressing

CIDR generalizes subnet addressing:

The Internet's address assignment strategy — $a.b.c.d/x$ where x bits form the NETWORK PREFIX

$a.b.c.d / x$ x leading bits = NETWORK PREFIX $32-x$ bits = HOST PORTION

CIDR in Practice

Organizations get a CONTIGUOUS BLOCK of addresses sharing a common prefix.

Example: organization owns $200.23.18.0/23$

→ $2^9 = 512$ consecutive addresses

→ All share the same $/23$ prefix

One forwarding entry covers all 512 addresses.

CIDR enables ROUTE AGGREGATION — keeps the global routing table manageable.

Route Aggregation — Textbook Example

ISP "Fly-By-Night" owns $200.23.16.0/20$.

Sub-allocates to 8 organizations:

Org 0: $200.23.16.0/23$

Org 1: $200.23.18.0/23$

...

Org 7: $200.23.30.0/23$

The ENTIRE INTERNET uses ONE entry:

$200.23.16.0/20$ → Fly-By-Night

External routers never see the 8 internal subnets. This is ADDRESS AGGREGATION.

Route aggregation: a single $/20$ entry covers $8 \times /23$ organizations → dramatic reduction in global routing table size

Classful Addressing vs CIDR — Why It Changed

Section 4.3.2: IPv4 Addressing

Before CIDR (before 1993): only three allowed prefix lengths — A=/8, B=/16, C=/24

Class	Prefix	Hosts per block	Total blocks	Problem
A	/8	16,777,214	128	Way too large — only large ISPs/gov
B	/16	65,534	16,384	Too large — 2,000-host org wastes 63,534 addresses
C	/24	254	2,097,152	Too small — can't host a medium company

Example: A company with 2,000 hosts needs Class B (/16 = 65,534 addresses) → wastes 63,534 addresses that no other organization can use!

CIDR Fix (1993): same company gets a /21 block (2,046 host addresses) — waste reduced to just 46 addresses

Address allocation hierarchy: ICANN → Regional Internet Registries (ARIN, RIPE, APNIC, LACNIC, AFRINIC) → ISPs → Organizations

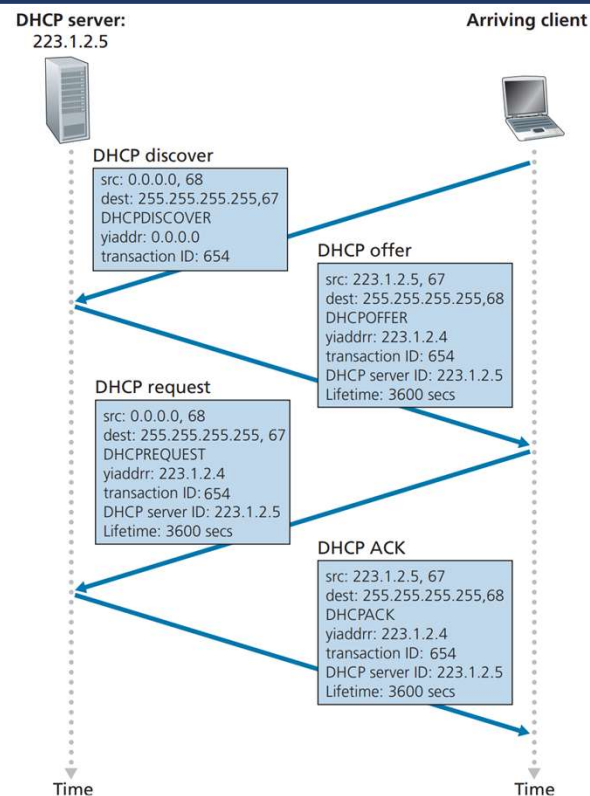
Private address blocks (RFC 1918) — NOT globally routable, reused by millions of networks:

10.0.0.0/8	16.7 million addresses — large enterprise
172.16.0.0/12	1.05 million addresses — medium networks
192.168.0.0/16	65,536 addresses — used in virtually EVERY home router

DHCP — Dynamic Host Configuration Protocol

Section 4.3.2: IPv4 Addressing

DHCP = PLUG-AND-PLAY (zero-conf) — a newly connected host obtains everything it needs automatically



DHCP shortcoming: new IP each subnet → TCP connections can't survive network handoff as mobile node moves between subnets

DHCP — Dynamic Host Configuration Protocol

Section 4.3.2: IPv4 Addressing

What DHCP provides: IP address · Subnet mask · Default gateway (router) · Local DNS server address

1 DHCP DISCOVER — finding a server

New host has NO IP. Broadcasts: Source = 0.0.0.0, Destination = 255.255.255.255, port 67. Reaches all devices on subnet including DHCP server.

2 DHCP OFFER — server responds

DHCP server broadcasts an offer: "I offer you 223.1.2.4 for 1 hour." Also offers subnet mask, default gateway, DNS server. Broadcast because client has no address yet.

3 DHCP REQUEST — client accepts

Client broadcasts acceptance: "I accept your offer of 223.1.2.4." Still broadcast — multiple servers may have offered, this tells all of them which offer was accepted.

4 DHCP ACK — confirmed

Server confirms: "Confirmed — 223.1.2.4 is yours for 1 hour." Client now sends and receives IP datagrams. Lease renewable before expiry.

DHCP — Operation Details and Deployment

Section 4.3.2: IPv4 Addressing

DHCP is widely deployed in residential, enterprise, and wireless networks — any environment with frequent host join/leave

DHCP Technical Details

DHCP is a CLIENT-SERVER protocol.

Each subnet has a DHCP server OR a router acting as DHCP RELAY AGENT forwarding requests to a centralized server.

DHCP Discover: yiaddr field = 0.0.0.0 ("this host")

DHCP Offer: yiaddr = proposed IP address

DHCP Request: client echoes back offered address

DHCP ACK: server confirms the lease

LEASE DURATION: typically 1 hour to 24 hours. After expiry, client must renew or stop using the address. Allows address reuse.

DHCP in Your Home Network

Your home router is BOTH a DHCP client (toward your ISP) AND a DHCP server (toward your devices):

1. Router contacts ISP's DHCP → gets public IP (e.g., 82.45.16.8)
2. Router runs its own DHCP server for home devices
3. Your laptop → Router DHCP → gets 192.168.1.101
4. Your phone → Router DHCP → gets 192.168.1.102
5. Your TV, printer, etc. all get private addresses automatically

The router also uses NAT (Section 4.3.3) to allow all these devices to share the single public IP.

Special IPv4 addresses:

0.0.0.0	"This host" — source address during DHCP discover before an address is known
255.255.255.255	Broadcast to all hosts on the same subnet
127.0.0.1	Loopback — packets never leave the host (local testing)

4.3.3 NAT — Network Address Translation

Section 4.3.3: NAT

An entire home or enterprise network shares a SINGLE public IP address using port number mapping

The Problem NAT Solves

IPv4 has only 4.3 billion addresses — running out by the early 2000s.

A typical home has multiple devices but receives only ONE IP from the ISP.

Without NAT: each device needs a public IP — impossible to provide.

With NAT: the entire home looks like ONE device to the outside world. All outgoing traffic uses the router's public IP as source.

How NAT Is Set Up

NAT router has:

- PRIVATE interface (home): 10.0.0.1
- PUBLIC interface (ISP): 138.76.29.7

Home devices get private addresses from router's DHCP: 10.0.0.1, 10.0.0.2...

The 10.0.0.0/8 block (RFC 1918) cannot be routed on the public Internet.

NAT router maintains a TRANSLATION TABLE:
(private IP, private port) ↔ (public IP, public port).

Port numbers are 16 bits → one public IP address can support OVER 60,000 simultaneous connections from different inside hosts

NAT — Translation Table and Step-by-Step

Section 4.3.3: NAT

The NAT translation table maps (WAN IP : WAN port) ↔ (LAN IP : LAN port)

Detailed Example (Figure 4.25): Host 10.0.0.1 → Web server 128.119.40.186:80

- 1 Host 10.0.0.1 assigns source port 3345 (arbitrary). Sends: Src=10.0.0.1:3345 → Dst=128.119.40.186:80
- 2 NAT router intercepts. Rewrites: Src=138.76.29.7:5001 → Dst=128.119.40.186:80.
Records in NAT table: (138.76.29.7:5001) ↔ (10.0.0.1:3345). Forwards to Internet.
- 3 Web server responds to 138.76.29.7:5001 (unaware of NAT). Reply: Src=128.119.40.186:80 → Dst=138.76.29.7:5001
- 4 NAT router looks up port 5001 in NAT table → finds 10.0.0.1:3345. Rewrites: Dst=10.0.0.1:3345. Delivers to correct inside host.

NAT Translation Table

WAN side address	LAN side address
138.76.29.7 : 5001	10.0.0.1 : 3345
138.76.29.7 : 5002	10.0.0.1 : 3346
138.76.29.7 : 5003	10.0.0.2 : 7532
138.76.29.7 : 5004	10.0.0.3 : 2210
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NAT — Widespread Use and Controversy

Section 4.3.3: NAT

NAT solves IPv4 exhaustion pragmatically — but violates fundamental Internet architecture principles

Why NAT Is Widely Deployed

Immediate, practical solution to IPv4 exhaustion — deployed in virtually EVERY home and small-office router worldwide.

No need to request or manage IP addresses for each device. Home users never configure IP addresses manually.

Provides implicit SECURITY: external parties cannot initiate connections to inside hosts (no entry in NAT table).

Allows easy renumbering: change ISP → only one public IP changes, inside network unchanged.

60,000+ simultaneous connections per public IP address.

Why NAT Is Controversial

PORT NUMBERS were designed to address PROCESSES, not hosts. NAT repurposes them to address devices — a misuse of the transport layer.

SERVICES behind NAT are unreachable from outside (no pre-existing NAT table entry). Complicates P2P applications, VoIP, gaming.

Layering violation: ROUTERS should process packets only up to the network layer (IP). NAT reads and modifies TRANSPORT-LAYER port numbers.

BREAKS END-TO-END CONNECTIVITY — one of the Internet's founding principles. The Internet was designed so hosts talk DIRECTLY to each other.

Workarounds exist but are complex: STUN, ICE, UPnP, NAT traversal (RFC 5389).

The PERMANENT solution to address exhaustion is IPv6 — not NAT. But NAT has kept IPv4 alive for 25+ additional years.

4.3.4 IPv6 — Why It Was Necessary

Section 4.3.4: IPv6

IPv4 has $2^{32} \approx 4.3$ billion addresses — the Internet exhausted them faster than anyone predicted

IPv4 Address Exhaustion Timeline

1994	IPv6 standardization process begins (RFC 2460: 1998)
1993	CIDR introduced to slow address depletion
2011 Feb	IANA global IPv4 pool EXHAUSTED
2011 Apr	APNIC (Asia-Pacific) runs out
2012 Sep	RIPE NCC (Europe) runs out
2015 Sep	ARIN (North America) runs out
2019+	RIPE enters strict rationing mode

IPv6 — The Permanent Solution

128-bit addresses:

$2^{128} \approx 3.4 \times 10^{38}$ addresses.

"Every grain of sand on the planet can be IP-addressable — and then some."

IPv6 introduces THREE key improvements:

1. EXPANDED ADDRESSING:
32 bits → 128 bits. Address exhaustion solved permanently.
2. STREAMLINED FIXED 40-BYTE HEADER:
Several costly IPv4 fields removed.
Fixed-length → faster router processing.
3. FLOW LABELING:
20-bit Flow Label field for traffic engineering and QoS.

Current deployment (~2024):

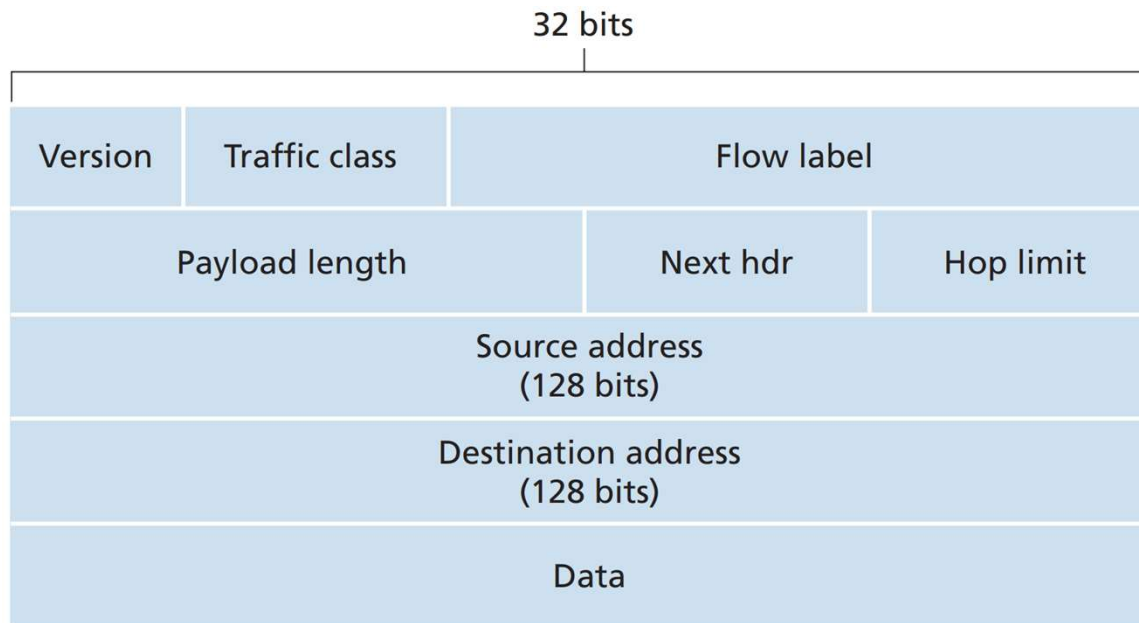
~40–45% of Google traffic served over IPv6.

Most modern devices are dual-stack (IPv4 + IPv6).

IPv6 Datagram Format — Fixed 40-Byte Header

Section 4.3.4: IPv6

IPv6 has a **FIXED 40-byte header** — no options, no fragmentation, no checksum — simpler and faster



Total: exactly 40 bytes — always. Data payload starts at byte 40, no variable-length fields to parse.

IPv6 Datagram Format — Fixed 40-Byte Header

Section 4.3.4: IPv6

Field	Size	Description
Version	4 bits	IP version = 6. (Putting 4 here does NOT create a valid IPv4 datagram.)
Traffic Class	8 bits	Priority marking — equivalent to TOS in IPv4. Used for DiffServ, ECN.
Flow Label	20 bits	NEW: identifies datagrams belonging to the same flow (video stream, VoIP call). Enables QoS.
Payload Length	16 bits	Number of bytes following the fixed 40-byte header (unsigned integer).
Next Header	8 bits	Identifies the next protocol: TCP (6), UDP (17), or an extension header. Same values as IPv4 Protocol field.
Hop Limit	8 bits	Decrement by 1 per router. Dropped at 0. Analogous to IPv4 TTL.
Source Address	128 bits	IPv6 address of sender. Written as 8 groups of 4 hex digits: 2001:db8::1
Destination Address	128 bits	IPv6 address of intended recipient.

IPv6 vs. IPv4 — What Was Removed and Why

Section 4.3.4: IPv6

Three costly IPv4 mechanisms removed from IPv6 — each removal speeds up router processing

Fragmentation Removed

In IPv4: routers can fragment large datagrams; only destination reassembles.

In IPv6: if a datagram is too large, the router DROPS it and sends a "Packet Too Big" ICMPv6 error to the source. Source retransmits with a smaller size.

Why remove? Fragmentation requires significant per-packet processing at every router. Moving it to the source speeds up IPv6 forwarding.

Header Checksum Removed

In IPv4: 16-bit 1's complement checksum over header. Must be RECOMPUTED at EVERY ROUTER because TTL decrements each hop.

In IPv6: No header checksum.

Why remove? At Gbps speeds, recomputing per-hop was prohibitively expensive. Error checking already done by TCP/UDP (transport) and Ethernet (link layer). The redundancy was unnecessary.

Options Removed

In IPv4: variable-length options → header has variable length → routers must find where data starts each hop.

In IPv6: options become EXTENSION HEADERS pointed to by the Next Header field. Completely optional — don't affect basic forwarding.

Why remove? Variable-length headers → unpredictable processing time at routers. Fixed 40-byte header = predictable, fast per-hop forwarding.

Net effect: IPv6 routers forward packets faster than IPv4 routers — simpler header, no recomputation, fixed-length parsing

IPv6 Addressing — 128-bit Notation and Types

Section 4.3.4: IPv6

$2^{128} \approx 3.4 \times 10^{38}$ addresses — enough for every grain of sand on Earth to have billions of addresses

IPv6 Address Notation

Eight groups of 4 hex digits, separated by colons:
2001:0db8:0000:0000:0000:ff00:0042:8329

SHORTENING RULES:

1. Leading zeros in a group may be omitted:
0042 → 42
2. One run of consecutive all-zero groups → ::
(:: may appear at most ONCE per address)

Shortened form:

2001:db8::ff00:42:8329

Special addresses:

::1 loopback (≡ IPv4's 127.0.0.1)
fe80::/10 link-local addresses
::ffff:192.168.1.1 IPv4-mapped IPv6

IPv4 vs IPv6 — Side by Side

	IPv4	IPv6
Address size	32 bits	128 bits
Total addresses	~4.3 billion	$\sim 3.4 \times 10^{38}$
Header size	20 bytes min	40 bytes fixed
Fragmentation	At routers + src	Source only
Header checksum	Yes (recomputed)	NONE
Options	Variable in hdr	Extension headers
Flow label	None	20-bit field
Notation	Dotted decimal	Hex with colons
Deployment	~60% of traffic	~40% of traffic

IPv4 → IPv6 Transition: Tunneling

Section 4.3.4: IPv6

A "flag day" flip-switch for billions of devices is impossible — tunneling enables gradual coexistence

Why Not a Flag Day?

When the Internet transitioned from NCP to TCP in 1983 it was already considered very difficult — and the Internet then was tiny and managed by a small number of experts.

With BILLIONS of devices today a simultaneous flag-day switch is completely unthinkable.

IPv6-capable systems can be backward-compatible (dual-stack), but IPv4-only systems cannot handle IPv6 datagrams.

Solution adopted: TUNNELING — the dominant transition mechanism.

How Tunneling Works

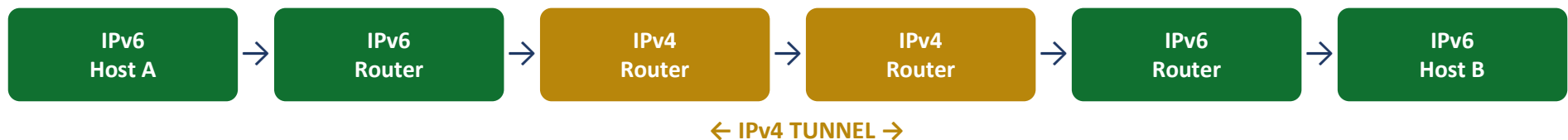
When two IPv6 nodes communicate through an IPv4-only region:

1. IPv6 datagram is placed as the DATA PAYLOAD of an IPv4 datagram.
2. IPv4 header: Src = tunnel entry, Dst = tunnel exit.
3. IPv4 routers inside the tunnel forward it normally — they see only IPv4.
4. At the tunnel exit, the IPv4 envelope is stripped off.
5. Original IPv6 datagram continues its journey.

From the IPv6 perspective: the IPv4 tunnel looks just like another link. IPv4 effectively acts as a LINK-LAYER PROTOCOL for IPv6!

Variants: 6to4, Teredo, DS-Lite, MAP-E.

[IPv6 datagram] → Tunnel Entry → [IPv4 [IPv6 datagram]] → Tunnel Exit → [IPv6 datagram]



Section 4.3 — Complete Summary Table

Section 4.3: Summary

IP is the core protocol of the Internet — understanding it fully is understanding the Internet itself

Topic	IPv4	IPv6 / New Concept
Header	20 bytes min, variable (options)	40 bytes FIXED — always
Addresses	32 bits · ~4.3 billion	128 bits · 3.4×10^{38}
Fragmentation	At routers + destination	Source only — routers drop+notify
Header checksum	Yes — recomputed at every router	REMOVED (too expensive per hop)
Options	Variable length in header	Extension headers (optional)
Flow identification	None	20-bit Flow Label
Address assignment	DHCP (4-step discover/offer/req/ack)	DHCPv6 or SLAAC (auto-config)
Address shortage	Fixed with NAT (short-term workaround)	Solved permanently by 128-bit space
Transition	Coexists via dual-stack and tunneling	IPv6-in-IPv4 tunneling
Current use (~2024)	~50% of Internet traffic	~50% of Internet traffic

Summary

4.3 — IP: IPv4, Addressing, NAT, IPv6

- IPv4 datagram: 20-byte header. Key fields: TTL (routing loops + Traceroute), Protocol (TCP=6, UDP=17 — demux key), Header Checksum (recomputed every hop — costly), fragmentation fields (destination only reassembles).
- IP address = interface address (not host). 32-bit, dotted-decimal. Subnet: set of interfaces with same prefix. Notation: 223.1.1.0/24 → 24-bit prefix, 254 hosts.
- CIDR (a.b.c.d/x): x-bit network prefix, 32-x host bits. Route aggregation: one /20 entry covers $8 \times /23$ organizations → manageable forwarding tables.
- Classful (A/B/C fixed 8/16/24) caused massive address waste → replaced by CIDR in 1993. Private RFC 1918 blocks: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- DHCP (plug-and-play): 4 steps — Discover (broadcast 255.255.255.255) → Offer → Request → ACK. Provides IP, subnet mask, default gateway, DNS server.
- NAT: entire home network shares one public IP via port number mapping. NAT table: (public IP:port) ↔ (private IP:port). 60,000+ connections per public IP. Controversy: violates layering, breaks end-to-end, complicates P2P/servers.
- IPv6: 128-bit addresses ($2^{128} \approx 3.4 \times 10^{38}$) — permanent fix. Addresses exhausted: IANA 2011, APNIC 2011, RIPE 2012, ARIN 2015.
- IPv6 header: FIXED 40 bytes. NEW: 20-bit Flow Label. REMOVED: fragmentation (source responsible), header checksum (too expensive per hop), options (become extension headers).
- IPv6 notation: 8 groups of 4 hex digits separated by colons. Compression: leading zeros omitted, :: replaces consecutive zero groups.
- IPv4→IPv6 transition via TUNNELING: IPv6 datagram encapsulated as IPv4 payload. IPv4 acts as link-layer protocol for IPv6. Current: ~40% of Internet traffic is IPv6.